MALLA REDDY TECHNICAL CAMPUS
A Constituent Unit of MALLA REDDY VISHWAVIDYAPEETH
Deemed to be University
Approved by UGC & AICTE New Delhi
Maisammaguda, Medchal (Dist), Hyderabad - 500100, Telangana

Department of Computer Science and Engineering

ALGORITHMIC PROBLEM-SOLVING HACKATHON (APSH 2026)

PROJECT DOCUMENTATION

Date: 10-10-2026 | Venue: 2nd Floor, A & B Blocks

Field	Details
Project Title	Blood Donor Finder
Team Name / Team No.	4
Year / Semester / Section	II Year B.Tech / 1st Sem /F
Algorithm Paradigm Used	Scalable Backend Architectures / microservices design and deployment/ database optimization/ high throughput API engineering/ cloud native systems and devops pipelines
Programming Language / Tools	Visual Studio Code/ Node.js/ React/ MongoDB Atlas/ Express.js
Faculty Mentor	Alekhya

Team Members

S.No	Name of the Student	Roll No.	Role
1	L. Pranav	25MVCSMR0506	Team Lead
2	L. Ganesh	25MVCSMR0505	Team member
3	G. Pannagesh	25MVCSMR0486	Team member
4	T. Praval	25MVCSMR0531	Team member

1. Problem Statement

Develop a MERN-based blood donor finder that allows users to find potential donors by blood group and location. The system should manage donor name, blood group, location, and contact information in MongoDB. React.js should provide the frontend, while Node.js and Express.js should expose backend APIs for creating, viewing, updating, searching, and managing donor records.

2. Objectives

- Demonstrate full-stack web development using the MERN stack
- Store and manage donor records using MongoDB.
- Implement CRUD operations through RESTful APIs.
- Build a responsive and user-friendly interface with React.js.
- Provide search and filtering by blood group and location
- Add authentication and role-based authorization where required.
- Present project information and results in a clear, accessible format.
- Protect personal data and encourage donor consent and accurate information.

3. Proposed Approach

The MERN-Based Blood Donor Finder is a web application designed to help users locate potential blood donors according to blood group and location. The system uses MongoDB to store donor information, Express.js and Node.js to provide backend REST APIs, and React.js to deliver an interactive user interface. The application is planned to support donor registration, viewing donor records, searching and filtering, updating details, and removing or deactivating records when appropriate. Authentication and role-based access can be used to protect donor data and administrative functions. The purpose is to make donor information easier to organize and search while emphasizing consent, privacy, and responsible use of contact details

4. Algorithm / Pseudocode

Step 1: Start the application.

Step 2: Connect the Node.js and Express.js backend to the MongoDB database.

Step 3: Display the React frontend with options to register, log in, search donors, and manage donor details.

Step 4: Allow the user to select a blood group and enter a location.

Step 5: Send the search request from React to the Express REST API.

Step 6: Retrieve donor records from MongoDB that match the selected blood group and location.

Step 7: Display matching donor details on the React interface.

Step 8: Allow authorized users to add, view, update, or delete donor records.

Step 9: Validate user input and apply authentication and authorization where required.

Step 10: Display the result or an appropriate error message.

Step 11: Stop the operation or continue accepting user requests

5. Complexity Analysis

Give the time and space complexity, with justification.

Complexity	Best Case	Average Case	Worst Case
Time	$O(1)$	$O(1)$	(O(N)\)*
Space	$O(N)$	$O(N)$	$O(N)$

6. Implementation

The Blood Donor Finder is implemented using the MERN stack: MongoDB, Express.js, React.js, and Node.js. The system allows users to register donor details, search for donors by blood group and location, view records, and manage donor information through CRUD operations

7. Results and Test Cases

Test case	Expected result	Status
Register with valid donor details	Record is saved and confirmation is shown	To be tested
Submit a missing required field	Validation message appears; invalid record is not saved	To be tested
Search by blood group	Only matching eligible records are returned	To be tested
Search by blood group and location	Results match both selected filters	To be tested
Update a donor record	Authorized changes are saved and displayed	To be tested
Attempt unauthorized update	API rejects the request	To be tested
Deactivate a donor	Inactive record is excluded from normal search	To be tested
Search with no matches	A clear no-results message is displayed	To be tested

8. Conclusion and Future Scope

The MERN-Based Blood Donor Finder is a practical full-stack development project that combines a React interface, Node.js and Express REST APIs, and MongoDB data storage. Its main functions are donor registration, search by blood group and location, and controlled management of records. The project provides a useful setting to demonstrate CRUD operations, API design, frontend-backend integration, and authentication. Privacy, donor consent, and careful wording about medical suitability must remain central to the design.

9. References

• MongoDB Documentation — https://www.mongodb.com/docs/
• Express.js Documentation — https://expressjs.com/
• React Documentation — https://react.dev/
• Node.js Documentation — https://nodejs.org/docs/
• Mongoose Documentation — https://mongoosejs.com/docs/